

Does change depend on baseline value?

Trait heritability¹

Plasticity
trade-offs^{2,3,4}

Compensatory
growth or
metabolism^{5,6}

Telomere length^{7,8}

Population size⁹

Species richness¹⁰

Range expansion¹¹

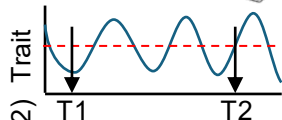
Ecological
disturbance¹²

Sources of variance in change

Baseline effect

Change depends on initial value (+, -, or 0)

Non-repeatability (or "measurement error")



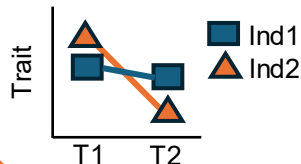
Instrument
error

Fluctuating
trait values



Non-heritable
variance

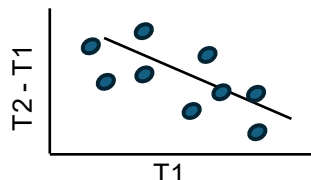
Heterogeneity in change



G x E

Context
dependence

Regression of change on baseline



Observed slope reflects
baseline effect and *RTM bias*

Methods to correct RTM bias

1. Estimate repeatability
(Blomqvist 1977)
*Targets *RTM bias* only*

2. Assume no
heterogeneity in change
(Kelly & Price 2005)
*Can mistake *change variance* for *RTM bias* and
overcorrect*